

MAT 2440: Data Structures and Algorithms

Fall 2026 – Section 610D

Instructor: Prof. Patricia Medina

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Class time and location: 8:00 am – 9:40 am, N-923

Office hours: 9:45 am – 10:40 am

Credits: 4

Main programming language: Python

Main communication platform: Brightspace

Description: Topics include fundamental data structures and algorithms, algorithm analysis, recursion, arrays, linked lists, stacks, queues, trees, heaps, hash tables, graphs, searching, sorting, and algorithm design. Students will implement and analyze algorithms using Python.

Co/Prerequisite: MAT 1375 or higher and one of the following: CST 1201 or CST 2403 or MAT 1630.

Components of Student's Grade

Evaluation	Percentage
3 In-Class Exams (16% each)	48%
Mini-Project(s) with presentation	14% (1 or 2)
Class Activities/Labs	8%
Final Exam	30%
Total	100%

Tentative Exam Schedule

Assessment	Date	Coverage
Exam 1	September 28, 2026	Logic, predicates, quantifiers, rules of inference, and proofs
Exam 2	October 26, 2026	Sets, functions, sequences, and algorithms
Exam 3	November 23, 2026	Algorithm analysis, complexity, number theory, and cryptography
Final Exam	December 21, 2026	Comprehensive final examination

Exam dates are tentative and subject to change.

Grading Scale

93–100	A
90–92	A-
87–89	B+
83–86	B
80–82	B-
77–79	C+
70–76	C
60–69	D
0–59	F

Attendance Policy

- Attendance of every session is mandatory. Please be on time. Lateness and students not staying for a complete class will be recorded. Two of these are equivalent to one absence. If a student has more than three absences, it may result in a WU grade.
- Excused absences must be arranged in advance of the session whenever possible, and work must be made up promptly on an agreed-upon schedule.
- Help is available during my office hours. Do not wait until the night before an exam to get help. If there is something you do not understand, seek help immediately.

Academic Integrity Policy

For the CUNY policies and definitions on cheating and plagiarism, see the College Catalog.

Students With Disabilities

The Center for Student Accessibility is the disabilities service provider at New York City College of Technology. The Center provides support to enrolled students who have documented permanent or temporary disabilities. The Center provides assistance in the areas of enrollment, academic advisement, tutoring, assistive technologies, and testing accommodations.

The Center is located in room L-237. To schedule an appointment, call 718-260-5143 or email Accessibility@citytech.cuny.edu.

Expectations and Behavior

- Expect to maintain an environment for learning. I request that cell phones and other electronic devices are silenced. Please also keep any other distractions to a minimum. In particular, no texting during class.
- Most importantly, each person should expect to maintain and appreciate an environment of respect and dignity. We will work together to gain understanding and ensure the success of every person.
- You are expected to be here each and every class. That means actively participating. We have a limited number of minutes. Let's make good use of them.
- Read the assigned textbook material and come to class prepared to work with classmates.
- Programming assignments, class activities, and labs should be completed carefully and submitted by the stated deadlines. Students are expected to write, test, and explain their own code.
- Be organized. Keep all of your classwork and programming assignments.
- No make-up exams are given without documentation. A make-up exam is offered only for valid absences, such as jury duty or illness. For illness, please submit documentation directly to the Student Accessibility Office. For deaths in the family, please go to the Office of Student Affairs, N-325.
- The final exam will be used to replace an exam missed for an excused absence.
- If you have difficulties, please seek help. You are welcome to come to my office hours. No appointment is needed.

- Read your CUNY email and check Brightspace regularly.
- Students will use Python throughout the course.

Writing Intensive Requirement

MAT 2440 is a Writing Intensive (WI) course. Students are expected to produce written work throughout the semester. Written work may include mathematical explanations, proofs, programming assignments, project reports, learning logs, and other forms of written communication.

Because of the increasing use of generative AI tools, some writing activities may be completed in class to provide students with opportunities to demonstrate their own understanding and mathematical reasoning.

Programming

Students will use Python throughout the course. Programming assignments and labs will be used to implement and analyze algorithms and to reinforce course concepts.

Course Resources

Course Outline: [City Tech MAT 2440 Course Outline](#)

Final Exam Review: [MAT 2440 Final Exam Review](#)

Additional project ideas and faculty resources are available through the MAT 2440/2540 faculty resources repository on OpenLab.

Textbook

The official textbook for Fall 2026 is:

Kenneth H. Rosen, *Discrete Mathematics and Its Applications*, 2025 Edition, McGraw-Hill.

The 8th edition and 7th edition may also be used. Problem numbers and page numbers may differ between editions.

Optional Meetings

Optional meetings may be held during the semester to discuss course content, answer questions, and exchange ideas about the course. Information about these meetings will be announced through Brightspace.

Tentative Course Schedule

Class	Topics
1	Propositional Logic: propositions, logical connectives, truth tables, and compound propositions
2	Applications of Propositional Logic: translating statements, logic puzzles, and Boolean expressions
3	Propositional Equivalences: logical equivalences, De Morgan's laws, and simplifying propositions

4	Propositional Equivalences: equivalence laws and applications; review of logical reasoning
5	Predicates and Quantifiers: predicates, universal and existential quantifiers, and translating mathematical statements
6	Nested Quantifiers and Rules of Inference: quantified statements and valid arguments
7	Introduction to Proofs: direct proofs, indirect proofs, proof by contradiction, and proof writing
8	Exam 1
9	Sets: set notation, subsets, cardinality, power sets, and Cartesian products
10	Set Operations: union, intersection, difference, complement, and set identities
11	Functions: definitions, domain and range, one-to-one and onto functions, and inverse functions
12	Sequences and Summations: sequences, recurrence notation, summations, and applications
13	Algorithms: algorithm specification, pseudocode, basic algorithmic problem solving, and examples
14	Algorithms: searching algorithms, linear search, binary search, and algorithm correctness
15	Algorithms: sorting algorithms, algorithm comparison, and introduction to efficiency
16	Exam 2
17	Growth of Functions: rate of growth, Big-O notation, and comparing functions
18	Growth of Functions: Big-O, Big-Omega, Big-Theta, and asymptotic comparisons
19	Complexity of Algorithms: analyzing running time and number of operations
20	Complexity of Algorithms: best-case, worst-case, and average-case analysis
21	Complexity of Algorithms: analyzing pseudocode and determining algorithm efficiency
22	Divisibility and Modular Arithmetic: divisibility, congruences, and modular operations
23	Integer Representations and Algorithms; Primes and Greatest Common Divisors: Euclidean algorithm and prime numbers
24	Solving Congruences and Applications of Congruences: linear congruences and modular arithmetic applications
25	Cryptography: basic cryptographic concepts, Caesar cipher, modular arithmetic, and applications
26	Exam 3
27	Mathematical Induction: principle of mathematical induction and basic induction proofs
28	Mathematical Induction: strong induction and applications to sequences, sums, and algorithms
29	Review: comprehensive review of logic, sets, functions, algorithms, complexity, number theory, and induction
30	Final Examination
